TEN-FRAMES & PLACE VALUE ARITHMETIC LAB

5-Anchor Subitizing, Regrouping & Base-10 Manipulatives

SERIES 1 • CARD 01 / 12 • DOMAIN: PLACE VALUE & 5-ANCHOR SUBITIZING

CARD 01 / 12 • BASE-10

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

5-ANCHOR SUBITIZING & BRIDGING TO 10

Ten-frames visually organize numbers into 5-anchors:

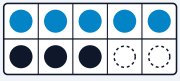
5-Anchor Formula: 5 + n • Ten's Comp: 10 - n

Seeing 8 as 5 on top and 3 below anchors addition,
turning counting into instant visual recognition!

CALIBRATED 14.00 MM SUBITIZING GRID

- Counter Diameter: 14.00 mm (2U) with two-tone face
- Stacking Datum: 1.85mm base plug & stacking socket
- Ten-Frame Grid: 2 × 5 array with 14.00 mm wells

5-ANCHOR SUBITIZING: 5 + 3 = 8 (BRIDGE 10)



INSTANT SUBITIZING

Top Row: 5 Blue (anchor)
Bottom: 3 Red (8 total)

2 Empty Slots:

10 - 8 = 2 Ten's Comp
Bridge: 8 + 5 = 13

2. KIT PARTS INVENTORY (56 PIECES)

COMPONENT	QTY	SPEC	COLOR
10-Frame Trays	2x	97.4×38.6mm	PLA Gray
Counter Blocks A	20x	19.0 mm Tile	PLA White
Landmark [10] A	6x	19.0 mm Tile	PLA White
Counter Blocks B	20x	19.0 mm Tile	PLA Tan
Landmark [10] B	6x	19.0 mm Tile	PLA Tan
Twin Stage Lid	1x	Twin 10-Frame	PLA White
3x1 Storage Bin	1x	49.0 mm	PLA White

TOTAL: 40 Counters + 12 Landmark [10] Blocks + 2 Trays + Box + Lid = 56 Parts

SLANT 3D PRODUCTION PALETTE

- PLA White: Set A (20x Counters + 6x [10] Blocks), Box Body & Stage Lid.
- PLA Tan: Set B (20x Counters + 6x [10] Blocks) for second operand.
- PLA Gray: Dual 5x2 Ten-Frame activity trays for high contrast.
- All 52 blocks pack in 4 columns of 13 blocks + 2 side trays.

3. PACK-UP & SUPPORT

✓ **10-SECOND PACK-UP**

Populate both 10-frames with 10 counters each and lay carry cubes into center trough. Empty wells instantly spot missing pieces. Twin stage lid snaps flush.

✓ **NO LOST PIECE GUARANTEE**

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ **HANDS-ON TUTORIAL**

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Base-10 positional notation decomposes any integer into powers of 10 with digits 0 through 9.

LESSON GUIDE: MAKE-A-TEN BRIDGE & 5-ANCHOR SUBITIZING

Step-by-Step Hands-On Tutorial: 8 + 5 = 13 via Bridging to 10

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LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/ten-frames/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 1.OA.C.6 (Make Ten)
- CCSS 2.NBT.B.5 (Place Value)
- TEKS 111.3.b.3.C (Compose/Deco...)

Grades K–3 Subitizing & Place Value

GUIDING INQUIRY

How many empty spots are left in Frame A before you start moving counters from Frame B?

1 STEP 1: POPULATE 8 ON TEN-FRAME A

Place 8 blue counters in Ten-Frame A.
Fill the top row (5 blue) and bottom row (3 blue).
Instant Subitizing: 5 + 3 = 8 (zero counting needed!).
Notice: exactly 2 slots remain empty to

2 STEP 2: POPULATE 5 ON TEN-FRAME B

Place 5 red counters in Ten-Frame B (fills top row of 5).
We want to calculate 8 + 5 without counting by ones.
Frame A needs 2 counters to bridge to 10.

3–4 STEPS 3 & 4: BRIDGE TO 10 & REGROUP

3. BRIDGE TO 10: Move 2 red counters from B to A. Frame A is now 1 Ten (10)!

4. READ REMAINING ONES: Frame B has 3 red counters left. Total: 10 + 3 = 13!

Keystrokes: Ten-Frame Icon ([▤] & [PV]) 8 [ENTER] 2 [+] 5 [+] [▤] → 15.

5 CORE PRINCIPLE: ADDITION IS BRIDGING, NOT COUNTING

Counting on fingers by ones is slow and error-prone. The 10-frame proves that addition is decomposing numbers into friendly anchors: 8 + 5 = 8 + (2 + 3) = (8 + 2) + 3 = 10 + 3 = 13! Once students see the 2 empty spaces in the 8-frame, they instantly know to borrow 2 from the 5!

★ EXTENSION INVESTIGATION: BRIDGE TO 10: 7 + 6

Put 7 in Frame A (3 empty). Put 6 in Frame B. Move 3 to make a ten. How many are left in B? 10 + 3 = 13! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.